PYTHON QUIZ GAME

1. PROJECT TITLE

Python Quiz Game Using Random Module

2. AIM

To develop a Python Quiz Game that displays multiple-choice questions, accepts answers from the user, checks the answers, and calculates the final score.

3. INTRODUCTION

The Python Quiz Game is a simple mini project developed using Python programming. It contains multiple-choice questions with four options for each question. The user selects an answer, and the program checks whether it is correct or incorrect. The final score is displayed at the end of the quiz.

The program uses the random module to shuffle the questions so that they appear in a different order each time the game runs.

4. OBJECTIVES

To develop an interactive quiz game using Python.

To display multiple-choice questions and options.

To accept and validate user answers.

To calculate the score automatically.

To understand lists, dictionaries, loops, conditional statements, and functions from Python's standard library.

5. SOFTWARE REQUIREMENTS

Programming Language: Python

IDE: Visual Studio Code

Module Used: Random

Operating System: Windows

6. ALGORITHM

Start the program.

Import the random module.

Store five questions, their options, and correct answers in a list of dictionaries.

Shuffle the questions using random.shuffle().

Initialize the score to zero.

Display each question and its four options.

Accept the user's answer and convert it to uppercase.

Compare the user's answer with the correct answer.

Display the result and increase the score if the answer is correct.

Display the final score and stop the program.

7. SOURCE CODE

```
import random

quiz = [
{
'question': 'What is the capital of France?',
'options': ['A. Berlin', 'B. Madrid', 'C. Paris', 'D. Rome'],
'answer': 'C'
},
{
'question': 'Which planet is known as the Red Planet?',
'options': ['A. Earth', 'B. Mars', 'C. Neptune', 'D. Saturn'],
'answer': 'B'
},
{
'question': 'What is the largest ocean on Earth?',
'options': ['A. Atlantic', 'B. Indian', 'C. Arctic', 'D. Pacific'],
'answer': 'D'
},
{
'question': 'How many days are there in a week?',
'options': ['A. 5', 'B. 6', 'C. 7', 'D. 8'],
'answer': 'C'
},
{
'question': 'Which of the following is a programming language?',
'options': ['A. Python', 'B. HTML', 'C. CSS', 'D. XML'],
'answer': 'A'
}
]

random.shuffle(quiz)

score = 0

for index, item in enumerate(quiz, 1):
print(f"Question {index}: {item['question']}")

for option in item['options']:
    print(option)

answer = input("Your answer (A/B/C/D): ").upper().strip()
```

```
if answer == item['answer']:
    print("Correct!")
    score += 1
else:
    print(f"Wrong! The correct answer is {item['answer']}.")

print()
```

print(f"Quiz over! Your final score is {score} out of {len(quiz)}.")

8. SAMPLE OUTPUT

Question 1: How many days are there in a week?
A. 5
B. 6
C. 7
D. 8

Your answer (A/B/C/D): C
Correct!

Question 2: What is the capital of France?
A. Berlin
B. Madrid
C. Paris
D. Rome

Your answer (A/B/C/D): C
Correct!

Question 3: Which planet is known as the Red Planet?
A. Earth
B. Mars
C. Neptune
D. Saturn

Your answer (A/B/C/D): B
Correct!

Question 4: What is the largest ocean on Earth?
A. Atlantic
B. Indian
C. Arctic
D. Pacific

Your answer (A/B/C/D): D
Correct!

Question 5: Which of the following is a programming language?
A. Python
B. HTML
C. CSS
D. XML

Your answer (A/B/C/D): A
Correct!

Quiz over! Your final score is 5 out of 5.

Note: The actual question order may differ because the questions are shuffled randomly.

9. CONCEPTS USED

List: Stores all the quiz questions.

Dictionary: Stores each question, its options, and its correct answer.

Random Module: Shuffles the questions randomly.

For Loop: Iterates through the questions and options.

If-Else Statement: Checks whether the user's answer is correct.

Input Function: Accepts the user's answer.

Score Variable: Tracks the number of correct answers.

Enumerate Function: Generates question numbers.

String Methods: upper() converts the answer to uppercase, and strip() removes extra spaces.

10. ADVANTAGES

Easy to understand and use.

Provides immediate feedback.

Calculates the final score automatically.

Displays questions in random order.

Helps beginners practise Python programming concepts.

11. CONCLUSION

The Python Quiz Game was successfully developed using basic Python programming concepts. It displays multiple-choice questions, accepts answers from the user, checks their correctness, and calculates the final score. This mini project helps improve knowledge of Python lists, dictionaries, loops, conditional statements, and the random module.